

Fertility-Household Block: One-Page Equation Summary

The only essential math update is to separate the old single child state into:

$$h_t = \text{children currently at home}, \quad p_t = \text{children ever born}.$$

The model household is one homogeneous fertility household. The housing crowding channel uses h_t , while completed fertility moments are matched with p_t .

State.

$$x_t = (b_t, a_t, z_t, h_t, p_t, t),$$

where b_t is debt/assets, a_t is owned housing, z_t is idiosyncratic income, $h_t \in \{0, 1, 2, 3\}$, and $p_t \in \{0, 1, 2, 3+\}$.

Flow utility without a birth.

$$u_t^N = \log c_t + s_h \log \tilde{H}_t + \phi_h h_t, \quad \tilde{H}_t = \frac{H_t}{(1 + \lambda h_t)^\psi},$$

where H_t is housing services, λ is the crowding strength, and ψ is the crowding exponent.

Flow utility with a birth.

$$u_t^B = \log c_t + s_h \log \tilde{H}_t^B + \phi_h h_t + \phi_p(p_t) - \kappa_b - \eta_q q_t, \\ \tilde{H}_t^B = \frac{H_t}{(1 + \lambda(h_t + 1))^\psi},$$

where $\phi_p(p_t)$ is the parity-specific birth utility, κ_b is the birth cost, and $\eta_q q_t$ is the housing-price birth penalty.

Budget constraint.

$$c_t = y_t(z_t, t) + R(b_t)b_t + q_t a_t - q_t a_{t+1} [1 + \kappa_a \mathbf{1}\{a_{t+1} \neq a_t\}] + b_t - b_{t+1},$$

with a rental branch when $a_{t+1} = 0$.

Bellman problem.

$$V_t(x_t) = \sigma \log \left(e^{V_t^N(x_t)/\sigma} + e^{V_t^B(x_t)/\sigma} \right), \\ V_t^N(x_t) = \max_{a_{t+1}, b_{t+1}} \left\{ u_t^N + \beta \mathbb{E}_t[V_{t+1}(x_{t+1}^N)] \right\}, \\ V_t^B(x_t) = \max_{a_{t+1}, b_{t+1}} \left\{ u_t^B + \beta \mathbb{E}_t[V_{t+1}(x_{t+1}^B)] \right\}.$$

Birth choice probability.

$$\Pr(d_t = 1 \mid x_t) = \frac{1}{1 + \exp(-(V_t^B(x_t) - V_t^N(x_t))/\sigma)}.$$

Child-state transitions.

$$p_{t+1} = \min\{p_t + d_t, \bar{p}\}, \\ h_{t+1} = \min\{\max(h_t + d_t - L_{t+1}, 0), \bar{h}\}, \quad L_{t+1} \sim \text{Bernoulli}(\pi_t).$$

The leave-home hazard π_t is reduced-form and is calibrated from leave-by-age bins (20, 25, 30 years) rather than from completed fertility moments.

Calibration mapping.

$$p_t \longrightarrow \text{target moments } (0, 1, 2, 3+) \text{ ever born}, \\ h_t \longrightarrow \text{housing crowding and children-at-home diagnostics}.$$

Current main parity target at age 50:

$$(0.165, 0.193, 0.357, 0.285).$$

Political equilibrium condition.

$$\Delta V(x_t; q) = V_t(x_t; q^+) - V_t(x_t; q), \quad \text{pref}(x_t; q) = \text{sign}(\Delta V(x_t; q)), \\ \text{TotalVote}(q) = \int \text{pref}(x_t; q) d\Gamma_t(x_t), \quad q^* : \text{TotalVote}(q^*) = 0.$$

Bottom line. The math change needed for the paper is modest: replace the old single child stock with (p_t, h_t) , make birth utility parity-specific through p_t , and let housing crowding depend on h_t only.